Nicola Dell'Avvocato

Mission Analysis · Systems Engineering · Space Systems

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PROFILE

MSc Space Engineering student working towards **mission analysis** and **systems engineering** for space missions, with a BSc in Aerospace Engineering completed at 110/110 cum laude. Hands-on with trajectory and flight-dynamics simulation (RocketPy, Poliastro, OpenRocket), space-debris risk assessment (ESA DRAMA) and Monte Carlo analysis, alongside requirements, subsystem interfaces and verification in a multidisciplinary engineering team (Capella, IBM DOORS). Comfortable working in English and Spanish in international environments, and looking for a career on European space programmes.

TECHNICAL SKILLS

Mission analysis & flight dynamics: RocketPy, Poliastro, OpenRocket, OpenVSP, Monte Carlo simulation, ESA DRAMA (space debris risk assessment and mitigation), Mission Planner (UAV ground control)

Systems engineering: Capella (MBSE), IBM DOORS (requirements management), requirements definition, subsystem interfaces, verification, multidisciplinary team coordination

Programming & numerical methods: MATLAB, Simulink, Python, digital signal processing (DSP), FFT and spectral analysis, data analysis, LaTeX

Structural simulation & CAD: Abaqus/CAE, finite element analysis (FEA), modal analysis, SolidWorks, CAD/CAM software, 3D modelling and additive manufacturing

Domains of interest: space missions and satellite constellations, CubeSats, Earth observation, space debris, dual-use and UAV technologies, AI applied to aerospace

Experimental & laboratory: CFRP composite lay-up, Fibre Bragg Grating (FBG) optical sensors and interrogators, impact-hammer modal testing, dynamic mechanical analysis (DMA), strain measurement, structural health monitoring, test-data post-processing

Transferable: team leadership and coordination, project management, analytical problem solving, critical thinking, interpersonal communication, work in international environments

EDUCATION

MSc in Space Engineering Sep 2026 – expected Jul 2028
Politecnico di Torino | Turin, Italy

BSc in Aerospace Engineering Sep 2023 – Jul 2026
Politecnico di Torino | Turin, Italy

- Final grade: 110/110 cum Laude.
- Erasmus+ exchange semester at Universidade da Coruña, Ferrol, Spain (Feb – Jun 2026).

› **Merit-based University College** — supplementary training programme, Collegio Universitario di Torino Renato Einaudi (Sep 2025 – Jul 2028)

› **Scientific high school diploma** — Cambridge International Curriculum, Liceo Scientifico Federico II di Svevia, Altamura — 100/100 (Sep 2018 – Jul 2023)

EXPERIENCE & ACTIVITIES

System Engineer — ICARUS PoliTo (student team) Oct 2026 – present
Politecnico di Torino | RA (Record Aircraft) solar-powered UAV | Turin, Italy

- System engineering on a solar-powered fixed-wing UAV designed for long-endurance flight on solar energy alone.
- Requirements definition, interfaces between subsystems and verification within a large multidisciplinary team.

Other work experience

› **Computer Laboratory Technician** — Politecnico di Torino — technical support in the University computer laboratories (part-time) (Sep 2026 – present)

› **Front Desk Receptionist** — Collegio Universitario Renato Einaudi — reception and access management (part-time) (Sep 2026 – present)

› **Staff** — Wave by Vento (formerly Italian Tech Week) — organising staff of an international technology conference (Jun – Oct 2026)

› **Head Lifeguard** — Tenuta Giannini, Sannicandro di Bari — pool safety and coordination of the lifeguard team (Jun – Sep 2023)

PROJECTS & RESEARCH

Challenge@PoliTO by Leonardo — Simulation & Experimental Validation (upcoming) Feb 2027 – Jun 2027
Politecnico di Torino & Leonardo | selected for a multidisciplinary company-led challenge | Turin, Italy

- Analysis and optimisation of electromagnetic and thermal signatures in complex physical systems.
- Development and validation of numerical models and experimental testing techniques, combining simulation, laboratory testing and data analysis in a real-world research environment.

CASSINI Hackathon — EU Space for Peace and Resilience (in progress) Nov 2026
European Commission / EUSPA | 12th edition

- Developing a team solution built on EU satellite data — Copernicus Earth observation and Galileo positioning — for a peace and resilience challenge.

F450 Quadcopter with AI Mission Interface (personal project, in progress) Oct 2026 – present
Personal project | additive manufacturing and autonomous flight

- Design and build of an F450-class quadcopter with additively manufactured structural parts, integrating airframe, avionics and software.
- Integration of an AI chat interface that translates natural-language input into autonomous mission configurations.

Embedded FBG Sensors for SHM of CFRP Sandwich Structures (BSc thesis) Jun 2026
Politecnico di Torino | Turin, Italy

- Built an Abaqus/CAE finite element model of a composite panel to predict its natural frequencies and strain mode shapes, with an optical fibre embedded between the carbon plies.
- Ran an impact-hammer test campaign with an optical interrogator and processed the strain signals through FFT in MATLAB, comparing measurements against the model.

Further projects

› **Dynamic Characterisation of 3D-Printed Metamaterials** — Universidade da Coruña — damping response vs. frequency on a DMA machine, for aeroelastic applications (Erasmus project) (Feb – May 2026)

› **INTRAPRENDENTI — Smart Infrastructures** — Politecnico di Torino Honour School — multidisciplinary team project with engineering, architecture and design (Sep 2024 – Jul 2025)

LANGUAGES, CERTIFICATIONS & HONOURS

Languages: Italian (native) · English, professional working proficiency · Spanish, limited working proficiency

Certifications: B2 First (English), Cambridge International Education · B2 DELE (Spanish), Universidade da Coruña · Cambridge International Curriculum IGCSE (A^* Chemistry, A^* Latin) · BLSD/AED, Croce Rossa Italiana · Lifeguard Certification, Federazione Italiana Nuoto